Private traits in digital records

EM1421 - Social media and web analytics

Slide courtesy of Prof. Fabiana Zollo

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Introduction

Kosinski et al., 2013, PNAS



Data classification in the digital age

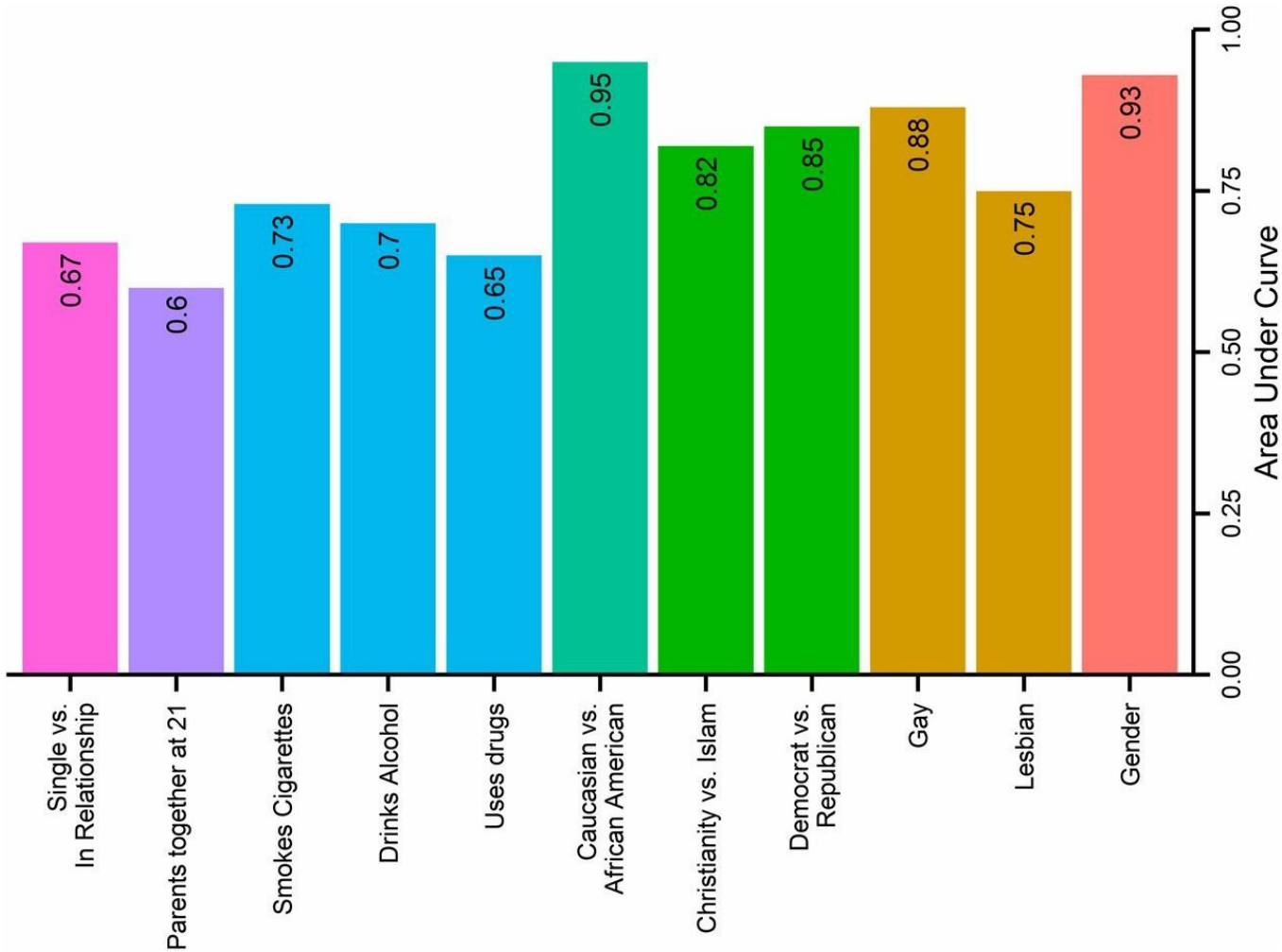
- *We show that easily accessible digital records of behavior, Facebook Likes, can be used to automatically and accurately predict a range of highly sensitive personal attributes including: sexual orientation, ethnicity, religious and political views, personality traits, intelligence, happiness, use of addictive substances, parental separation, age, and gender*
- In the digital age, data classification is not so obvious (e.g. personal and sensitive data)

The study

- Kosinski et al. study the liking behavior of 58K US Facebook users (avg 170 likes per user)
- They build a prediction model for the following variables:
  - Facebook profile: age, gender, political and religious views, relationship status, proxy for sexual orientation, social network size and density
  - Profile picture: ethnicity
  - Survey results: BIG5 Personality, intelligence, satisfaction with life, substance use, parents together?

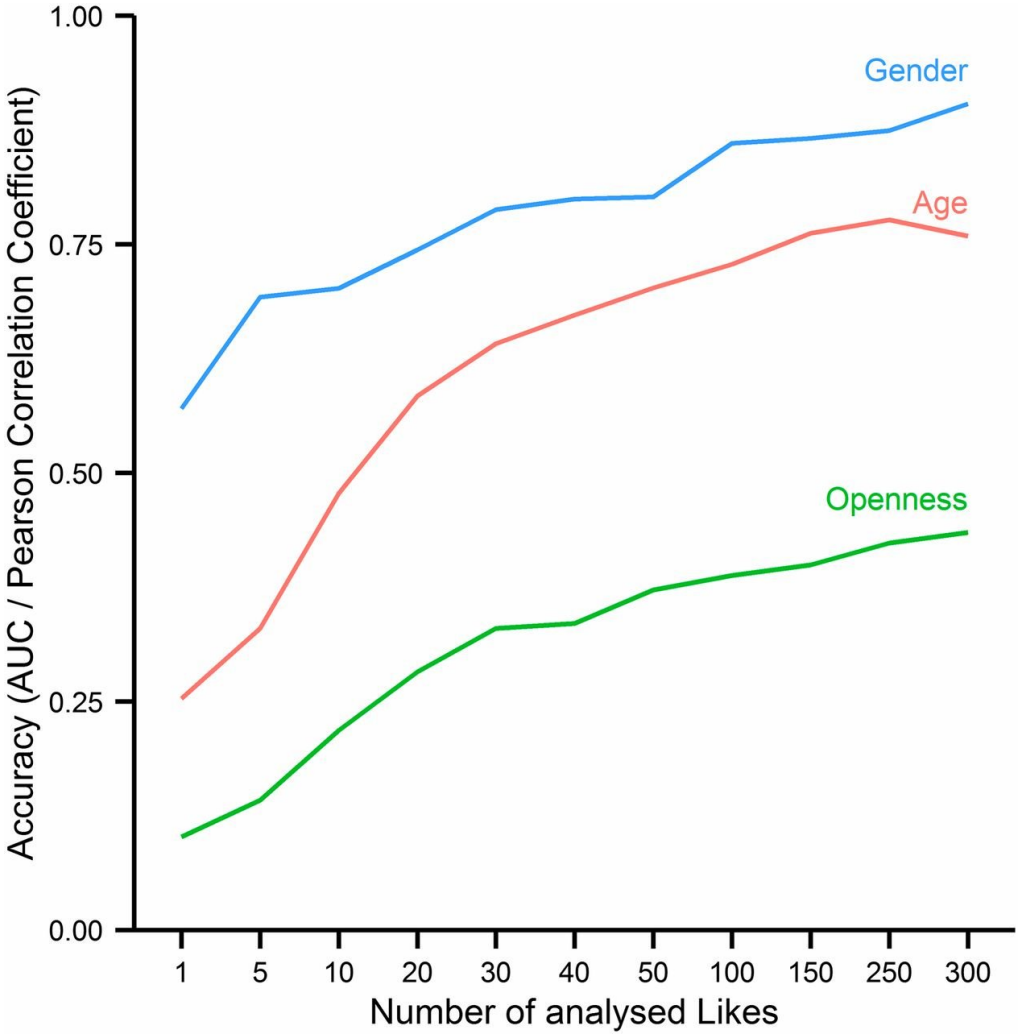
Results

Overall accuracy of predictions:



Results

Accuracy as a function of the number of likes in input:



Notes

This presentation is based on the paper: Kosinski, M., Stillwell, D., & Graepel, T. (2013). Private traits and attributes are predictable from digital records of human behavior. Proceedings of the national academy of sciences, 110(15), 5802-5805.